Year 12 Maths - Working with data

Similar questions - Scatter diagrams and correlation

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Eight students recorded the number of hours, x, they spent revising for a test and their mark, y (out of 100).

x	2	3	5	6	8	9	11	12
y	31	38	44	52	60	63	75	78

(a) Describe the correlation between x and y. **[1]**

(b) The line of best fit is $y = 22.5 + 4.66x$. Interpret the values 22.5 and 4.66 in context. **[2]**

(c) Estimate the mark for a student who revises for 7 hours, and explain why the same line should not be used to predict the mark for a student who revises for 30 hours. **[3]**

(d) A friend says: "This proves that revising for longer causes higher marks." Comment on this statement. **[2]**

2 The table shows, for six Local Authorities (LAs), the percentage x of employees who use public transport to travel to work and the percentage y who use a car, van or motorcycle.

LA	P	Q	R	S	T	U
x (%)	10	25	45	5	60	20
y (%)	80	65	45	82	30	30

(a) Show that five of the LAs lie close to the line $x + y = k$ for a suitable constant k, and state k. **[2]**

(b) Explain what k tells you about the employees who use neither of these two types of transport. **[2]**

(c) Identify the LA that is an outlier from this pattern and suggest a reason in context for its position. **[2]**

(d) Without calculation, describe the correlation you would expect between x and y for all the LAs in the UK, and explain why. **[2]**

Question 1

(a) Strong positive correlation

(b) 22.5: the predicted mark for a student who does no revision. 4.66: each extra hour of revision is associated with an increase of about 4.7 marks

(c) $22.5 + 4.66 \times 7 = 55.1$, so about 55. 30 hours is outside the range of the data (2 to 12 hours): extrapolation. The line would give $22.5 + 4.66 \times 30 = 162$, which is impossible as the maximum mark is 100

(d) Correlation does not prove causation. Other factors (for example prior ability or motivation) may affect both revision time and marks. The data suggest a link, but a controlled experiment would be needed to claim a cause

Question 2

(a) $x + y = 90, 90, 90, 87, 90, 50$ for P, Q, R, S, T, U respectively. Five are 90 (or 87, close), so $k = 90$

(b) About 10% of employees use neither: for example they walk, cycle, or work mainly at or from home

(c) U, since $x + y = 50$ is far from 90. About 50% use neither transport type, which suggests a rural or island area, or a small town where people walk to work or work from home

(d) Negative correlation: the more people use public transport, the fewer use cars, because the two percentages are limited to total about 90%